



Ford American Journey 2.0 (AJ)—a future model of Ford Fiesta

engineers exploring the possibilities of connecting an automobile to the Internet.

"It's getting pretty dark; time to put the headlights on," read a tweet by AJ when the team of engineers drove the car to California for an exhibition. Followers also learned that AJ's mood was 'joyful' when "there was no traffic, and it was not raining and it was enjoying a winding road."

The software behind AJ is an application called the Auto'matic Blog that taps into the available data on the car, including information on its location, speed, acceleration and braking. It gleans information from windshield wipers, steering input and GPS data and correlates it with live information culled from the Web.

Now touch and feel 3-D images

Researchers at the University of California, San Diego, have created a virtual reality device—called 'heads-up virtual reality' (HUVR)—that allows users to not only see a 3-D image but feel it too. The device couples a consumer 3-D HDTV panel with a half-silvered mirror to project any graphic

image onto the user's hands and/or into the space surrounding him. With his head position tracked to generate the correct perspective view, the user manoeuvres a touch-feedback (haptic) device to interact with

the generated image, literally 'touching' the angles of the image and contours as if it was a tangible 3-D object.

HUVR is well-suited to training and education in structural and mechanical engineering, archaeology and medicine. The device could be used, for example, to visualise and manipulate the 3-D image of a person's brain taken from an MRI, or an artefact too fragile or precious to be physically handled.

Make calls even when network is not available

Apple's iPhone 4 has been making international headlines for reception problems caused by its antenna design. But, what if a software lets you make and receive phone calls even when there is no network available? Well, Australian researchers have created such a software.

Researchers at Flinders University are testing a new technology that uses the Wi-Fi interface of mobile phones, to allow multiple mobiles to create their own, small network. It actually allows mobiles to transmit calls to one another without cell towers or satellites.

The software detects other phones in its Wi-Fi radius and places calls to them without needing an operator. Obviously, the big challenge for an idea like this is range. At best, a mobile phone will have a Wi-Fi radius of a few hundred metres. And that's in ideal circumstances—in a city or heavily urban area, the effective broadcast range will be much shorter. To fix this problem, the software allows your phone to act as a small cell tower. It means that your phone can pick up and relay someone else's phone call. So if there are other mobile devices in your area, your radius is much larger.

Super-quick charger for electric cars

A charger developed by a Japanese firm takes only three minutes for a 50 per cent charge, or five minutes for a 70 per cent charge. The engineering firm, JFE Engineering, recently demonstrated the prototype at *Smart Grid Exhibition 2010* held in Tokyo.

This represents substantial progress in shortening the charging time as conventional equipment, even those claimed to be quick chargers, often need 30 minutes to charge a battery to about 80 per cent of its capacity.

The new charging system can also work with low-voltage power supplies, being able to suck 20 kW from a normal European 220V AC grid. It is also believed to be grid-friendly, having two batteries inside: One has a higher capacity and is able to store grid electricity during night time, and the other is a high-output battery, ensuring the fast three-minute charging (probably acting as a buffer between the grid and its bigger sister).

Ppresenting The New

BRIVURA

Daring brilliant performance

540 TVL OSD Camera Series

Selective BLC Off	HSBLC Off	Masking Off
Selective BLC On	HSBLC On	Masking On